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Topic 8 — Nuclear and Particle Physics

8.1 Atomic Structure

The atom consists of a small, dense, positively charged nucleus (containing protons and neutrons, collectively called nucleons) surrounded by orbiting electrons. The nucleus is approximately 10^{-15} m in diameter, while the atom is approximately 10^{-10} m — the nucleus is about 100,000 times smaller than the atom.

Particle	Relative Charge	Relative Mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	−1	1/1836 ≈ 0.0005	Orbits / electron cloud

Nuclear Notation

A nuclide is represented as: $^{A}_{Z}X$, where A = mass number (protons + neutrons), Z = atomic number (protons), and X is the chemical symbol. The number of neutrons is N = A − Z.

Isotopes

Isotopes are atoms of the same element (same number of protons, same Z) but with different numbers of neutrons (different A). They have the same chemical properties (determined by electrons) but different physical properties (e.g., different mass, different stability — some may be radioactive).

Worked Example: Carbon-12 ($^{12}_{6}C$) has 6 protons and 6 neutrons. Carbon-14 ($^{14}_{6}C$) has 6 protons and 8 neutrons. Both are carbon (Z = 6), but C-14 is radioactive and decays via beta-minus emission

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with a half-life of 5730 years, making it useful for radiocarbon dating.

Rutherford Scattering

The nuclear model of the atom was established by Rutherford's alpha-particle scattering experiment (1909-1911). When alpha particles were fired at thin gold foil:

- **Most passed straight through** → atoms are mostly empty space
- **Some were deflected at small angles** → the nucleus has positive charge
- **Very few bounced back (>90°)** → the nucleus is small, dense, and contains most of the atom's mass

8.2 Fundamental Particles

Quarks

Quarks are fundamental particles that make up hadrons. There are six flavours arranged in three generations. Only the first generation (up and down) makes up stable everyday matter.

Generation	Quark	Symbol	Charge	Baryon Number	Strangeness
1st	Up	u	+2/3	+1/3	0
1st	Down	d	−1/3	+1/3	0
2nd	Charm	c	+2/3	+1/3	0
2nd	Strange	s	−1/3	+1/3	−1
3rd	Top	t	+2/3	+1/3	0
3rd	Bottom	b	−1/3	+1/3	0

Each quark has a corresponding **antiquark** with opposite charge, baryon number, and strangeness. For example, the anti-up quark ($\bar{u}$) has charge −2/3 and baryon number −1/3.

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Exam Tip: Remember the charges: up-type quarks (u, c, t) have charge +2/3, and down-type quarks (d, s, b) have charge −1/3. Antiquarks have opposite signs for all quantum numbers.

Leptons

Leptons are fundamental particles that do not experience the strong nuclear force. They interact via the weak and electromagnetic forces (if charged).

Generation	Lepton	Symbol	Charge	Lepton Number (L_e, L_μ, L_τ)
1st	Electron	e^-	−1	$L_e = +1$
1st	Electron neutrino	ν_e	0	$L_e = +1$
2nd	Muon	μ^-	−1	$L_\mu = +1$
2nd	Muon neutrino	ν_μ	0	$L_\mu = +1$
3rd	Tau	τ^-	−1	$L_\tau = +1$
3rd	Tau neutrino	ν_τ	0	$L_\tau = +1$

Each lepton has a corresponding antilepton with opposite charge and lepton number. Neutrinos are extremely light (nearly massless), uncharged, and interact very weakly with matter — billions pass through your body every second from the Sun.

Exchange Particles (Gauge Bosons)

Exchange particles (gauge bosons) are virtual particles that mediate the fundamental forces between other particles. Each force has its own exchange particle(s).

Force	Exchange Particle	Acts On	Range	Relative Strength

Electromagnetic	Photon (γ)	All charged particles	Infinite	10^{-2}
Strong nuclear	Gluon (g)	Quarks and gluons	$\sim 10^{-15}$ m	1
Weak nuclear	W^+, W^-, Z^0	All fermions	$\sim 10^{-18}$ m	10^{-6}
Gravity	Graviton (hypothetical)	All particles with mass	Infinite	10^{-39}

The **W bosons** carry charge (W^+ has charge +1, W^- has charge −1) and are responsible for changing quark flavour in beta decay. The Z^0 boson is neutral and mediates neutral current interactions.

8.3 Hadrons

Hadrons are composite particles made of quarks, held together by the strong force. They are divided into two groups:

Baryons (3 quarks)

- **Proton (p):** uud — charge = +2/3 +2/3 −1/3 = +1, baryon number = +1
- **Neutron (n):** udd — charge = +2/3 −1/3 −1/3 = 0, baryon number = +1
- Baryon number = +1 for baryons, −1 for antibaryons
- The proton is the only stable baryon. Free neutrons decay via beta-minus with a half-life of about 10 minutes.

Worked Example: Verify the charge of the proton from its quark composition (uud).

Charge = (+2/3) + (+2/3) + (−1/3) = +4/3 − 1/3 = **+1** ✓

Baryon number = (+1/3) + (+1/3) + (+1/3) = **+1** ✓

Mesons (quark-antiquark pair)

- **Pion (π^+):** $u\bar{d}$ — charge = +2/3 +1/3 = +1
- **Pion (π^-):** $\bar{u}d$ — charge = −2/3 −1/3 = −1
- **Pion (π^0):** $u\bar{u}$ or $d\bar{d}$ — charge = 0
- **Kaon (K^+):** $u\bar{s}$ — contains an anti-strange quark, strangeness = +1
- **Kaon (K^-):** $\bar{u}s$ — strangeness = −1
- Baryon number = 0 for all mesons (quark B = +1/3, antiquark B = −1/3)

Pions are the lightest mesons and mediate the **residual strong force** between nucleons in the nucleus (also called the nuclear force). This is distinct from the fundamental strong force between quarks, which is mediated by gluons.

Antiparticles

Every particle has a corresponding **antiparticle** with the same mass but opposite charge, baryon number, lepton number, and strangeness. For example: the antiproton ($\bar{p}$) has quark composition $\bar{u}\bar{u}\bar{d}$ and charge −1.

> **Exam Tip:** When identifying quark compositions, remember: (1) baryons have three quarks, mesons have a quark-antiquark pair; (2) add up the charges to check they give the correct total; (3) the antiquark in a meson has the opposite signs for all quantum numbers.

8.4 Particle Interactions

Conservation Laws

In all particle interactions, certain quantities must be conserved. Whether a proposed reaction can occur depends on whether all conservation laws are satisfied.

Quantity	Conserved In
Charge (Q)	All interactions
Baryon number (B)	All interactions

Lepton number (L)	All interactions (separately for each generation: L_e, L_μ, L_τ)
Strangeness (S)	Strong and electromagnetic only (can change by 0 or ±1 in weak interactions)
Energy and momentum	All interactions

Worked Example: Determine whether the following reaction is allowed: $p + p \rightarrow p + p + p + \bar{p}$

Check charge: LHS = +1 + +1 = +2. RHS = +1 + +1 + +1 + (−1) = +2. ✓
Check baryon number: LHS = +1 + +1 = +2. RHS = +1 + +1 + +1 + (−1) = +2. ✓
Check lepton number: LHS = 0. RHS = 0. ✓

All quantities conserved, so this reaction **is allowed**. This is proton-antiproton pair production — it requires sufficient kinetic energy (at least $2m_pc^2$ = 1876 MeV).

Exam Tip: Strangeness is conserved in strong interactions but can change by ±1 in weak interactions. This is how strange particles are produced (in pairs via the strong force, e.g., $K^+ + K^-$) but decay individually via the weak force (relatively slowly).

Beta Decay at Quark Level

Beta-minus decay (β^-): A neutron decays into a proton, emitting an electron and an electron antineutrino.

$$n \rightarrow p + e^- + \bar{\nu}_e$$

At quark level: $d \rightarrow u + e^- + \bar{\nu}_e$ (mediated by W^- boson)

The down quark emits a W^- boson (changing to an up quark), and the W^- immediately decays into an electron and electron antineutrino.

Beta-plus decay (β^+): A proton decays into a neutron, emitting a positron and an electron neutrino.

$$p \rightarrow n + e^{+} + \nu_e$$

At quark level: $u \rightarrow d + e^{+} + \nu_e$ (mediated by W^{+} boson)

Electron capture: An inner orbital electron is captured by a proton in the nucleus:

$$p + e^{-} \rightarrow n + \nu_e$$

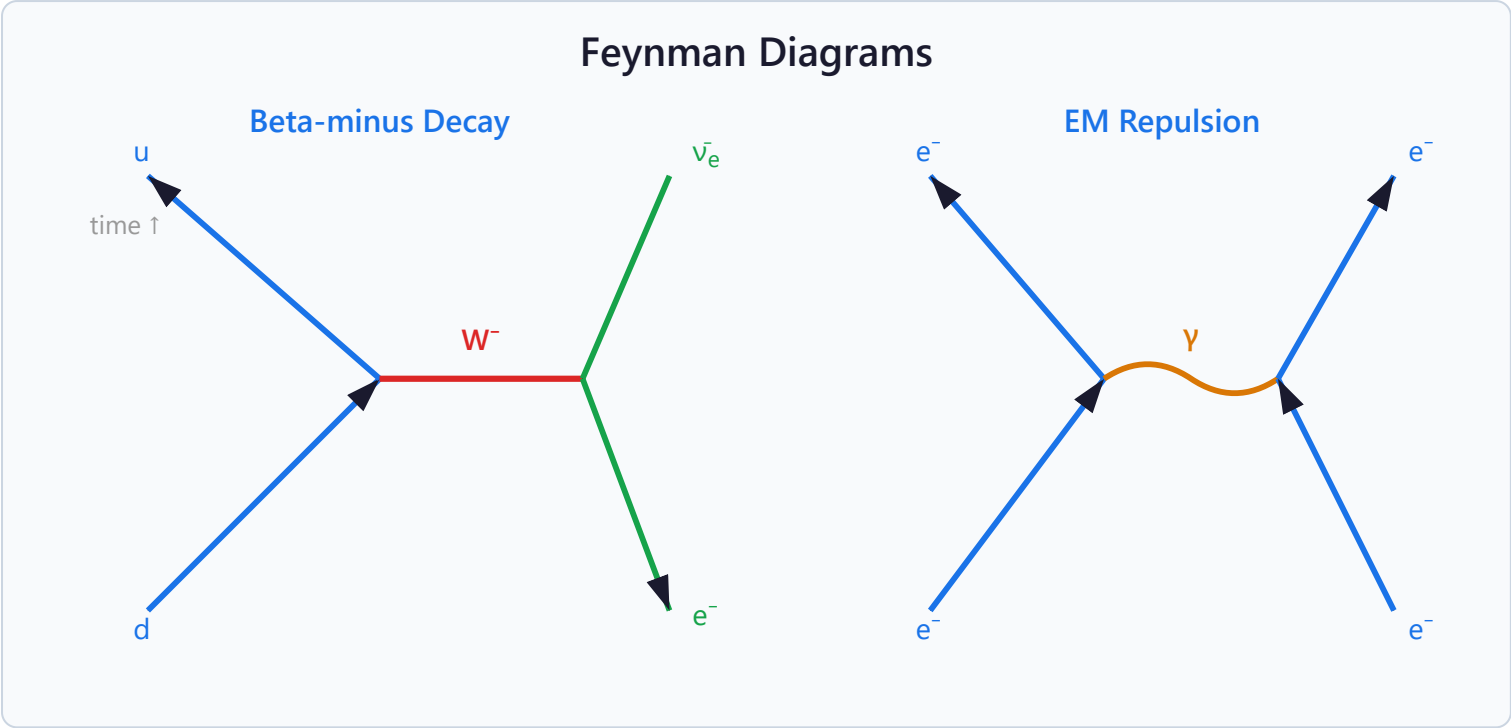
This is an alternative to beta-plus decay and is mediated by a W^{+} boson.

8.5 Feynman Diagrams

Feynman diagrams are spacetime diagrams that represent particle interactions. They are not literal pictures of what happens but are powerful tools for visualising and calculating interaction probabilities.

Convention:

- Time axis runs upward (or left to right — check the diagram)
- Straight lines represent fermions (particles/antiparticles)
- Wavy lines represent photons or W/Z bosons
- Curly lines represent gluons
- Antiparticles have arrows pointing backward in time
- At each vertex, charge, baryon number, and lepton number are conserved



Exam Tip: When drawing Feynman diagrams, always check conservation of charge at every vertex. The W^+ carries charge +1 and the W^- carries charge −1. The Z^0 and photon are neutral. At the quark-W vertex in beta-minus decay: charge in = −1/3 (d quark); charge out = +2/3 (u quark) + (−1) from W^-. Check: +2/3 − 1 = −1/3 ✓

8.6 The Standard Model

The **Standard Model** of particle physics classifies all known fundamental particles and describes three of the four fundamental forces (excluding gravity).

- **Fermions** (matter particles, spin ½): 6 quarks + 6 leptons (plus their antiparticles) = 24 matter particles
- **Bosons** (force carriers, integer spin): photon, $W^\pm$, Z^0, 8 gluons, Higgs boson = 13 force/field particles

The Standard Model of Particle Physics

Quarks (feel strong force)

u up +2/3	c charm +2/3	t top +2/3
d down -1/3	s strange -1/3	b bottom -1/3
Gen 1	Gen 2	Gen 3

Leptons (no strong force)

e electron -1	μ muon -1	τ tau -1
νe e-neutrino 0	νμ μ-neutrino 0	ντ τ-neutrino 0

Gauge Bosons (force carriers)

γ (EM) | g (Strong) | W±, Z^0 (Weak) | Graviton (?)

Scalar Boson

H (Higgs)

Quarks + Leptons = Fermions (matter) | Gauge + Higgs = Bosons (forces)
Hadrons = quarks bound by strong force: Baryons (3 quarks) + Mesons (quark-antiquark)

The **Higgs boson** was confirmed in 2012 at CERN. It is associated with the Higgs field, which gives particles their mass through interaction with the field. Particles that interact more strongly with the Higgs field have greater mass (the top quark interacts most strongly; neutrinos interact very weakly).

Limitations of the Standard Model

- Does not include gravity (no quantum theory of gravity yet exists)
- Does not explain dark matter or dark energy (which make up ~95% of the universe)
- Does not explain why there is more matter than antimatter in the universe
- Does not predict the masses of particles (these must be measured experimentally)
- Contains many free parameters (~19) that must be determined by experiment

8.7 Pair Production and Annihilation

Pair production: A photon with sufficient energy creates a particle-antiparticle pair. The minimum photon energy required equals the combined rest mass energy of the pair: $E = 2mc^2$.

$$\gamma \rightarrow e^- + e^+ \quad (\text{minimum photon energy} = 2 \times 0.511\ \text{MeV} = 1.022\ \text{MeV})$$

Pair production must occur near a nucleus (to conserve momentum — a single photon cannot produce a particle-antiparticle pair in empty space because momentum would not be conserved). The nucleus absorbs the recoil momentum.

Annihilation: When a particle meets its antiparticle, they annihilate, converting their combined mass-energy into photon energy. Two photons are produced (to conserve momentum).

$$e^- + e^+ \rightarrow 2\gamma$$

Each photon has minimum energy $E = mc^2 = 0.511$ MeV (for electron-positron annihilation at rest). The two photons travel in opposite directions to conserve momentum.

Worked Example: Calculate the minimum frequency of a photon that can produce an electron-positron pair.

Minimum energy = $2m_ec^2 = 2 \times 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13}$ J

$E = hf$, so $f = E/h = 1.64 \times 10^{-13} / 6.63 \times 10^{-34} =$ **2.47×10^{20} Hz**

This is in the gamma ray region of the electromagnetic spectrum.

Exam Tip: In annihilation, two photons are produced travelling in opposite directions to conserve momentum. In pair production, a nearby nucleus is needed to absorb recoil momentum. Always check that both energy and momentum are conserved.

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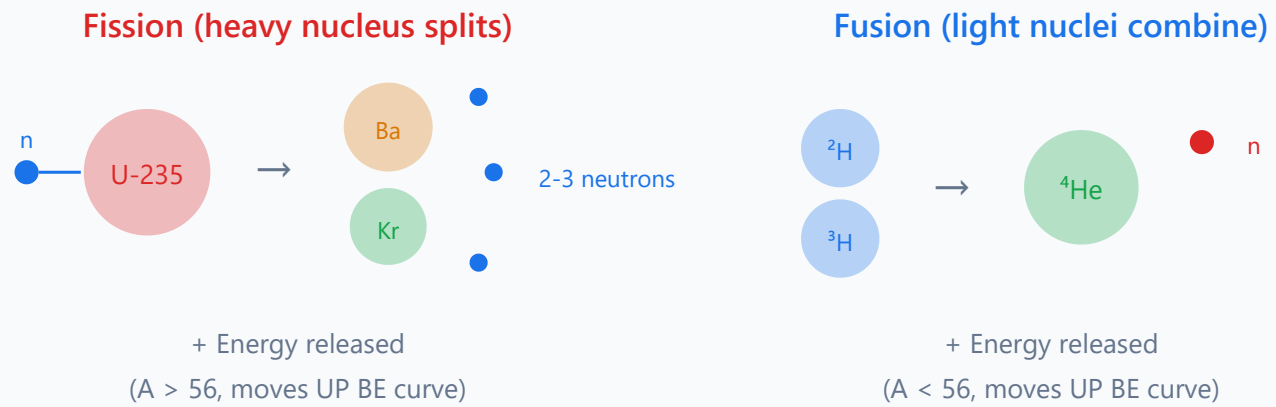
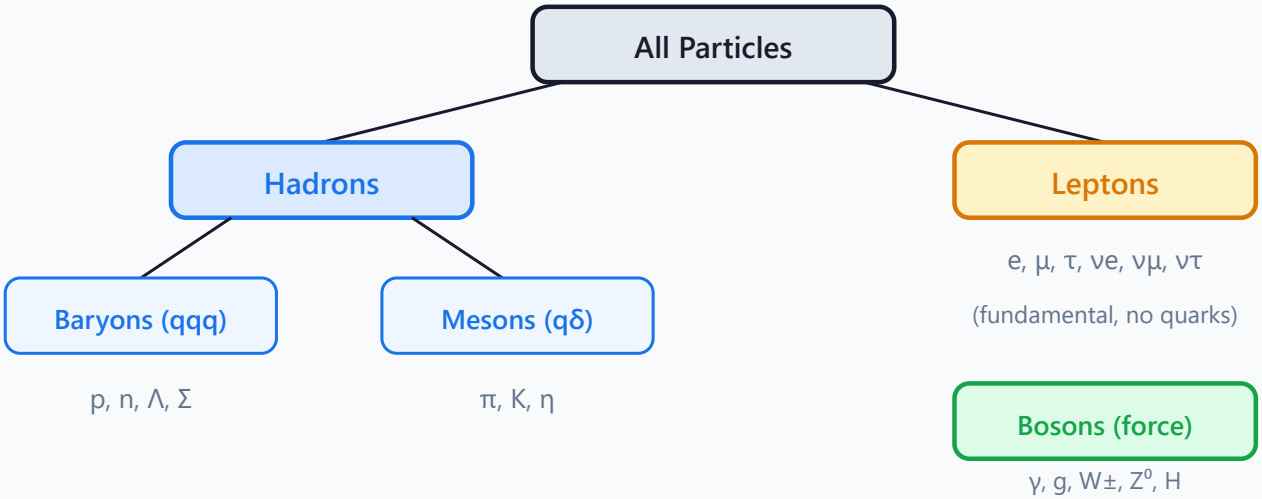
Nuclear Fission vs Fusion



8.8 Classification Summary

Particle Classification



Exam Tip: A common exam question asks you to classify a particle. The key decision tree is: (1) Does it feel the strong force? If yes → hadron. If no → lepton or boson. (2) If hadron: is it 3 quarks (baryon) or quark-antiquark (meson)? (3) Check quantum numbers add up correctly.